The AI Minister: Ethical and Engineering Implications of Automating the Ministry of Territory and Housing

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Abstract— This article explores the socio-technical and ethical implications of delegating the management of the Catalan Ministry of Territory and Housing to an Artificial Intelligence system, referred to as "Conseller-IA". The Ministry's core functions (urban planning, housing accessibility, and mobility) are analyzed as complex multi-objective optimization problems where technical efficiency often conflicts with social vulnerability and environmental sustainability. We present a rigorous technical framework that specifies the input feature sets, model architectures, and explainability mechanisms required for each sub-domain, alongside a quantified assessment of the system's material footprint, including energy and water consumption in the context of regional resource scarcity. Through three distinct case studies (urban renewal [1], emergency housing allocation, and energy infrastructure licensing) we evaluate the limits of algorithmic decision-making. The study concludes that while AI can enhance administrative efficiency, it cannot replace human phronesis in the face of ethical dilemmas, necessitating a "human-in-the-loop" governance model based on the principles of accountability, reversibility, and democratic representation.

Keywords— Artificial Intelligence, Algorithmic Governance, Public Administration, Tech Ethics, Multi-objective Optimization

I. Introduction

The rapid evolution of Large Language Models (LLMs) and predictive algorithms has catalyzed a digital transformation within public administrations worldwide. In the context of the Generalitat de Catalunya, the hypothetical implementation of a "Conseller-IA" represents the most ambitious expression of this trend: an automated system capable of managing a government department whose decisions directly affect the right to housing (Art. 47 CE) [2], territorial planning (Art. 9.9 EAC) [3], and the mobility of millions of citizens.

From an engineering perspective, the Ministry's tasks can be modeled as data-driven processes aiming for maximum efficiency over well-defined objective functions. However, delegating public power to an algorithm introduces critical challenges across three axes. First, the problem of Materiality: the physical infrastructure required for such a system entails significant energy and water costs, raising ethical concerns in a territory subject to severe drought protocols. Second, the Problem of Many Hands complicates accountability; when an automated decision results in a socio-economic error, the responsibility becomes dispersed among developers, data curators, and political supervisors in ways that resist clear attribution. Third, the Explicability Gap: the most performant models (deep neural networks, large ensemble methods) tend to produce opaque decision boundaries that are incompatible with the transparency obligations of public administration.

The central thesis of this work is that algorithmic governance, while superior in processing large volumes of structured data for logistical optimization, risks perpetuating historical biases and lacks the capacity for moral exception in cases of extreme human vulnerability. We structure our analysis by first defining the technical and material framework of the "Conseller-IA", specifying the variable sets and model types for each sub-domain, followed by an explicability analysis and a delegation matrix that establishes "red lines" for automation. We then apply four ethical models (Teleological, Deontological, Virtue Ethics, and Care Ethics) to three real-world scenarios to determine the limits of technical legality versus social justice.

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II. System Architecture and Technical Framework

a. The "Conseller-IA" Model Framework

The architecture of "Conseller-IA" must be conceived as a complex, hybrid socio-technical system rather than a monolithic algorithm. To effectively manage the Ministry of Territory and Housing, the system requires a multi-layered infrastructure that integrates both predictive and generative algorithmic paradigms, each operating over domain-specific feature sets.

1. Sub-system 1: Territorial Planning (Planificació Territorial)

Model architecture: Transformer-based LLMs for normative coherence analysis + Convolutional Neural Networks (CNN) for satellite image analysis. The predictive layer uses Gradient Boosting (XGBoost) for geological and flood risk classification.

The territorial planning sub-system ingests the following primary feature sets:

- Cartographic vulnerability maps (ICGC), including slope, land-use classification, and protected-area overlays (continuous raster, 1m resolution, ∼90 TB base layer).
- Demographic and socioeconomic indicators from IDESCAT [4]: population density, income quintiles, employment rate, and urbanization trends per census section (tabular, updated annually).
- Real-time and historical pollution sensor data from the Xarxa de Vigilància de la Contaminació: NOx, PM2.5, and CO_2 concentrations at fixed stations (time-series, 15-min intervals).
- Cadastral registry data: plot size, construction year, building permits, and current land classification (relational, ∼150 TB for all of Catalunya).
- Pirineu Strategy 2050 KPIs: biodiversity indices, sustainable mobility access scores, and agro-forestry zoning compliance (structured, updated quarterly).

2. Sub-system 2: Urban Planning & Building Control

Model architecture: Random Forest for habitability violation classification + Graph Neural Networks (GNNs) for neighbourhood-level connectivity and segregation analysis [5]. NLP pipelines (BERT-based) for automated processing of building inspection reports and legal filings.

Key input features:

- Full inspection records per building unit: ventilation measurements (m^3/h), lighting levels (lux), moisture readings, and Decree 141/2012 compliance flags (structured, ∼300 TB historical).
- POUM (Plan d'Ordenació Urbanística Municipal) [6] zoning vectors for all 947 Catalan municipalities: land-use polygons, floor-area ratios (FAR), height limits, and setback requirements.
- LiDAR point-cloud data for 3D urban morphology: building footprints, height profiles, and shadow analysis (unstructured, ∼80 TB, updated every 3 years).
- Citizen complaint records (text): OCR-extracted and NLP-classified by urgency and typology.
- Social vulnerability index (constructed feature): composite of income, household density, % elderly, % minors, and % non-EU residents per census section.

3. Sub-system 3: Housing Allocation & Emergency Response [7]

Model architecture: Multi-objective optimization (Pareto-front methods, specifically NSGA-II [8]) for housing resource allocation. Survival analysis models (Cox Proportional Hazards) for homelessness risk prediction. LLMs for automated drafting of official communications and legal documents.

Key input features:

- Official Housing Registry [9] (Registre de Sol·licitants d'Habitatge): waiting list entries, household composition, income certificates, current housing status (encrypted tabular, ∼40 TB).
- Real-time emergency reports: municipality-submitted crisis declarations, humanitarian organization reports (semi-structured).
- Climate-adjusted construction cost indices by municipality and construction type (tabular, updated quarterly).
- CO_2-equivalent emission factors per construction typology (prefabricated concrete, modular timber, passive-house standards) which is the critical trade-off variable in Case Study 2.

4. Sub-system 4: Mobility & Infrastructure Licensing

Model architecture: Reinforcement Learning (RL) agents for traffic flow optimization + Multi-Agent Systems (MAS) for public transport scheduling. Traditional ML classifiers for environmental impact screening.

Key input features:

- Real-time IoT traffic sensor data from the SCT (Servei Català de Trànsit): loop detector counts, average speeds, incident flags (streaming, ∼5 GB/day).

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- Public transport ridership data (FGC, TMB, Renfe Rodalies): O-D matrices, load factors, punctuality records (tabular, daily batch).
- Environmental impact reports for infrastructure projects: noise maps (ISO 9613-2 format), air quality dispersion models (Gaussian plume), hydrological impact assessments.
- Singular Landscape catalogue (PTOP) and ENPE network shapefiles: spatial polygons defining zones where development restrictions apply (the binding constraint in Case Study 3).

b. Data Infrastructure and Residency

All personal and sensitive data processed by the Conseller-IA must reside exclusively on sovereign infrastructure. The primary deployment is the Centre de Processament de Dades (CPD) of the Generalitat de Catalunya, located in Sant Cugat del Vallès, complemented by a secondary disaster-recovery node in Lleida. This dual-node architecture ensures compliance with GDPR Art. 44-49 (prohibition on transfers to third countries without adequate protection), the EU AI Act requirements for high-risk systems (Annex III, point 2), and the Catalan Government's digital sovereignty mandate.

No model inference may be routed through public cloud providers (AWS, Azure, GCP) for data classified above Level 2 in the Esquema Nacional de Seguretat (ENS). Anonymized, aggregated outputs (e.g., district-level housing demand forecasts) may be cached on federated cloud infrastructure to reduce latency for public-facing dashboards, provided that differential privacy guarantees ($\epsilon \leq 1.0$) are applied before export.

The total estimated data volume across all sub-systems is approximately 580–660 TB of structured and unstructured data, growing at an estimated 15% annually driven primarily by LiDAR updates and sensor time-series accumulation. A formal data retention policy is mandatory: personal data must be purged after the legally mandated conservation period, operationalizing GDPR's right to erasure and preventing indefinite accumulation that would inflate the environmental footprint of the system.

c. Material and Environmental Cost

The implementation of algorithmic governance introduces a significant, yet frequently ignored variable: the physical and environmental footprint of the AI itself. While cloud computing is often perceived as an intangible asset, the physical infrastructure required to sustain continuous inference and periodic retraining of the Conseller-IA demands immense physical resources. This creates a severe systemic paradox: an AI deployed to optimize territorial sustainability may inherently degrade the local environment through its own operation.

1. Energy Consumption Estimation

We estimate the annual energy consumption for each sub-system based on their computational profile (see Table 1).

The estimated total annual energy consumption of ∼3,600 MWh places the system in the range of a medium-sized municipal data center. To satisfy the Catalan Government's climate neutrality target (Llei 16/2017 del Canvi Climàtic [10]), 100% of this energy must be sourced from certified renewable generation (specifically, a Power Purchase Agreement (PPA) tied to wind or photovoltaic generation within Catalan territory). A carbon accounting dashboard must be integrated into the system's operational monitoring panel, displaying real-time carbon intensity per inference request.

2. Water Consumption Estimation

The thermal design power (TDP) of modern GPU clusters generates substantial heat dissipation, typically managed through evaporative cooling towers. In the context of Catalonia (a Mediterranean region subject to recurring drought emergencies and ACA emergency protocols) water consumption is not merely an operational cost but an ethical constraint.

For the Sant Cugat CPD with modern adiabatic free-cooling and hybrid wet/dry cooling systems, we estimate a Water Usage Effectiveness (WUE) ratio of approximately 0.8–1.2 L/kWh, detailed in Table 2.

An annual consumption of 2.9–4.3 million litres represents a non-trivial water burden in a region where agricultural and urban water demand regularly triggers restrictions. The engineering obligation is therefore to implement a conditional scaling policy: during ACA-declared drought phases 2 or 3, the Conseller-IA must automatically suspend all non-critical batch inference tasks (e.g., historical re-analyses, training cycles) and reduce active inference to essential operational sub-systems only, with a target WUE reduction of at least 40%.

3. Hardware Lifecycle and Geopolitical Dependencies

The rapid obsolescence of AI processing units (typically 3–5 year cycles for GPU generations) contributes to a growing stream of electronic waste. The procurement of GPU clusters relies on rare-earth elements and critical minerals, specifically neodymium and dysprosium (for permanent magnets in cooling systems), cobalt and lithium (for UPS battery systems), and tantalum (for capacitors in PCBs). Major extraction sites for these materials are concentrated in the Democratic Republic of Congo (cobalt), Chile and Argentina (lithium), and China (rare earths), representing geopolitical dependencies and documented cases of environmental degradation and labour rights violations in mining communities.

A truly ethical engineering approach requires that the procurement policy for Conseller-IA hardware

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Table 1: Estimated annual energy consumption by sub-system.

Sub-system	Hardware Profile	Training	Inference	Total (MWh/yr)
Territorial Pl.	8x H100 nodes	~320	~530	~850
Urban Pl.	4x H100 + 2x CPU	~180	~770	~950
Housing Alloc.	2x H100 + opt.	~95	~305	~400
Mobility	6x H100 + stream	~210	~1,190	~1,400
TOTAL		**~805**	**~2,795**	**~3,600**

Table 2: Water consumption estimation scenarios.

Scenario	Energy (MWh/yr)	WUE (L/kWh)	Water (ML/yr)
Conservative	3,600	0.8	~2.9
Base case	3,600	1.0	~3.6
Worst case	3,600	1.2	~4.3

mandates: (1) compliance with the EU Conflict Minerals Regulation (2017/821); (2) a minimum 5-year hardware lifecycle target with certified refurbishment before decommissioning; and (3) partnership with a certified WEEE (Waste Electrical and Electronic Equipment) recycler accredited under Catalan Law 15/2003. These constraints must be incorporated as hard requirements in the technical specification, not post-hoc sustainability add-ons.

III. Explicability and the Accountability Gap

a. The Engineering Requirement for Transparent Models

A critical dimension that distinguishes AI governance in the public sector from commercial applications is the obligation of explicability, which is the technical requirement that a decision-making system be interpretable both epistemologically (how the algorithm reaches a conclusion) and ethically (who is responsible for that conclusion). This dual requirement is mandated by the EU AI Act [11] (Art. 13, "Transparency and provision of information to deployers"), which classifies housing and territorial planning systems as high-risk AI under Annex III.

From a data engineering standpoint, the models with the highest predictive performance (deep neural networks, large ensemble forests, and LLMs) are precisely those that produce the most opaque decision boundaries. This creates a fundamental tension: optimizing for accuracy degrades explicability, and vice versa. The resolution of this tension is not optional for a public administration system; it is a hard constraint imposed by democratic accountability. We propose a layered explicability architecture tailored to each model type, as summarized in Table 3.

Critically, SHAP values [13] for the housing allocation and urban planning models must be generated at inference time, not post-hoc, and must be stored alongside every decision record in the audit log. This ensures that if a citizen or legal representative contests a decision, the technical basis of that decision can be reconstructed exactly as it appeared at the moment it was made, a requirement for due process under Spanish Administrative Law (Llei 39/2015, Art. 35).

b. The Problem of Many Hands and Accountability Engineering

The integration of Conseller-IA fundamentally disrupts the traditional chain of administrative responsibility. In a conventional bureaucratic framework, a flawed housing policy can be traced back to a specific civil servant or elected official. Algorithmic governance distributes this responsibility across: data engineers who curated the training datasets, ML engineers who designed the objective functions, third-party vendors supplying GPU infrastructure, and political supervisors who authorized deployment.

A common technocratic defense relies on the premise that technology is inherently neutral, that an algorithm is merely a tool and that any negative outcomes stem solely from its misuse. This argument is deeply flawed. Every mathematical regularization parameter applied to a predictive housing model, every feature included or excluded from the social vulnerability index, and every threshold set for the "high-risk" flag in the building inspection classifier constitutes a moral decision encoded into the system's architecture. Engineers developing algorithms for public administration are personally responsible to society and cannot disclaim the social repercussions of their work by deferring to political authorities.

To operationalize accountability, the engineering pipeline must enforce: a Model Card for each sub-system (documenting intended use, training data provenance, performance metrics disaggregated by demographic group, and known failure modes); a Data Statement for each training dataset; named technical and political custodians for each sub-system recorded in the audit trail; and an Algorithmic Impact Assessment (AIA) conducted before any major model update, modeled on the GDPR's Data Protection Impact Assessment (DPIA).

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Table 3: Explicability Techniques by Sub-system.

Sub-system / Model	Explicability Method	Output Format	Audit Level
Territorial Pl. (LLM)	Chain-of-thought + attention viz	Textual justification	Full text log
Urban Pl. (RF)	SHAP values per decision	Feature bar chart	Case-level report
Building Vio. (GNN)	GNNExplainer attribution [12]	Highlighted graph	Case-level report
Housing Alloc.	Pareto-front visualisation	Interactive chart	Decision-level log
Mobility RL Agent	Saliency maps + policy gradient	Heatmap of state-space	Aggregated weekly

IV. Delegation Matrix: Limits of Automation

a. *Taxonomy of Delegation Levels*

The delegation of authority requires a clear socio-technical taxonomy distinguishing between administrative optimization and political discretion. We propose a three-tier system:

- AI-Decision Maker (Full Delegation): Logistical optimization in closed systems with clearly defined, measurable objective functions and no direct impact on individual rights. Applicable to: real-time traffic signal optimization, public transport frequency scheduling, routing of non-urgent administrative correspondence, and anomaly detection alerts (flagging for human review). The algorithm is authorized to act autonomously only if the decision is fully reversible within 24 hours and affects no individual citizen's legal status.

- AI-Assistant (Supportive Automation): In the domains of urbanism and housing, the AI operates as a sophisticated diagnostic tool. The system processes massive datasets to detect residential anomalies (lack of ventilation, structural risks, normative non-compliance) and generates ranked recommendation lists for human review. The AI provides the empirical "what" and "where"; the human official retains the "why" and "how" of the intervention.

- Human-Only Decision (Hard Prohibition): Any decision that directly modifies an individual's legal status (eviction orders, demolition authorizations, housing allocation refusals, and infrastructure licensing) must require explicit validation from a human civil servant with qualified legal competence. The algorithm may not proceed autonomously regardless of the confidence score of its prediction.

b. *Systemic Red Lines*

The Moral Exception Problem: The algorithms operate through strict technical legality but lack human phronesis (practical wisdom). As the class debate made explicit, the solution to cases where normative compliance produces unjust outcomes is not to program the algorithm to autonomously deviate from the law, that would be a political and juridical contradiction. The Conseller-IA must execute legal compliance strictly; the responsibility for reforming inadequate laws rests exclusively with legislative human action. The algorithm flags conflicts between technical legality and social vulnerability for human escalation; it does not resolve them unilaterally.

Algorithmic Justice and Bias Persistence: Training on historical Generalitat data risks perpetuating past territorial inequalities. The social vulnerability index must be constructed using a documented, peer-reviewed methodology with annual external audits. Disaggregated performance metrics (precision, recall, and false-positive rates) must be computed across geographic regions, income quintiles, and household typologies. Any detected disparate impact above a predefined threshold triggers mandatory human review of the relevant model component.

Democratic Representativity: Under the deontological framework, an algorithm cannot hold the representative status of an elected official. The democratic legitimacy of Conseller-IA decisions derives entirely from the human official who validates each consequential output (not from the algorithm's statistical accuracy).

Data Neutrality and RLHF: The class debate explicitly excluded direct partisan political actors from the fine-tuning and RLHF process to prevent ideological bias injection into the model's core. The data engineer's role is not to achieve impossible absolute neutrality, but to exercise critical vigilance and active data curation (asymptotic impartiality) while documenting every curation decision in the audit trail.

V. Ethical Framework and Impact Analysis

a. *Philosophical Perspectives*

To rigorously evaluate the deployment of an algorithmic entity governing territorial and housing policies, we analyze its decision-making framework through four established ethical models. This multi-dimensional approach prevents a purely mathematical evaluation of the system and introduces necessary moral constraints.

- Teleological Ethics (Consequences): Rooted in utilitarianism, this model evaluates actions based on outcomes. For Conseller-IA, this translates to an objective function maximizing overall utility (optimizing transport efficiency or maximizing housing units built). However, a strictly teleological algorithm risks marginalizing statistical outliers, potentially sacrificing vulnerable minorities to achieve a mathematically optimal outcome for the majority. Hans Jonas's Imperative of Responsibility [14] adds a temporal dimension: the algorithm must weight the permanence of future human life (e.g., green corridors, flood-risk avoidance) even when this conflicts with immediate social urgency.

- Deontological Ethics (Principles): Following Kantian principles, this framework focuses on adherence to universal rules. An automated system is inherently deontological in execution, strictly applying urban codes without deviation. The engineering challenge arises when strict technical legality (Summum ius) conflicts with higher-order duties (such as protecting human dignity) where the algorithm's rigidity cannot process moral exceptions. As the class debate established, the ad legalem fallacy must be explicitly rejected: what is legal is not by definition morally correct.

- Virtue Ethics (Aristotle / Phronesis): Algorithms operate on deterministic or probabilistic mathematical models and lack practical wisdom. They cannot possess moral character or contextual intuition, making them incapable of the nuanced judgment required when urban policies severely impact human lives. The class notes emphasize that the data scientist bears the phronetic responsibility that the algorithm cannot exercise, including the obligation to audit outlier cases, as the algorithm cannot "give face" or explain the suffering it generates.

- Care Ethics (Tronto / Gilligan): This perspective centers on relationships, dependency, and vulnerability. Integrating Care Ethics into the system requires engineering Conseller-IA to mathematically weight human vulnerability as a primary constraint, not as an externality. A system that evicts a family for physical safety (ventilation) while ignoring their vital security (homelessness) acts from a technocratic and blind-to-suffering perspective. Per Joan Tronto [15], justice must recognize interdependence; the algorithm must be designed to halt when the care relationship is severed.

b. Begoña Román's Ten Ethical Values [16]

Table 4 details the application matrix of these values to the system.

c. Impact Hierarchy (Micro, Meso, Macro)

- Micro Level (The Individual Citizen): The introduction of algorithmic governance modifies the citizen's option sets and alters their perception of the administration. Carissa Véliz's analysis of privacy as power [17] is directly applicable: crossing datasets (water consumption + census registry + building inspection records) can convert public infrastructure into a panopticon that disproportionately surveils those who lack the economic resources to conceal their precarity. A high-efficiency system detecting illegality can become blind to the need that causes it, functioning as an audit tool for poverty rather than an instrument of social justice [18].

- Meso Level (The Administration): The deployment of Conseller-IA fundamentally modifies the distribution of power within the Generalitat. Civil servants transition from active decision-makers to algorithm supervisors, creating a risk of automation bias: routinely approving algorithm outputs due to workload pressure, which effectively transfers decision-making power back to the data engineers and historical datasets. This risk is not resolved by the human-in-the-loop model alone; it requires active organizational design interventions (structured override incentives and audit quotas for civil servant supervisors).

- Macro Level (Democracy and Algorithmic Justice): Algorithms trained on historical datasets risk perpetuating territorial and social inequalities. The Frankfurt School's critique of instrumental rationality applies directly: converting political questions (value judgments about spatial equity) into technical questions (optimization problems) depoliticizes decisions that are fundamentally democratic. If Conseller-IA allocates resources based on past investment patterns, it mathematically guarantees the persistence of infrastructural deficits in historically marginalized areas, encoding historical segregation into future predictive models.

VI. Case Studies: Algorithmic Optimization vs. Value Alignment

a. Empirical Validation: Stakeholder Value Alignment

To quantify the divergence in multi-objective optimization priorities, an empirical survey was conducted [19]. Respondents were asked to resolve 10 complex urban, social, and environmental dilemmas, acting as the decision engine for the "Conseller-IA". Crucially, the options provided for each dilemma corresponded to the core values of four distinct stakeholder archetypes: Ecological Conservation (A), Economic Efficiency (B), Left-wing Social Protection (C), and Real Estate/Property Rights (D). To prevent ideological bias, participants were blinded to this underlying categorization during the survey.

As illustrated in Figure 1, the results reveal a high degree of situational variance in algorithmic value

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Table 4: Begoña Román's Ten Ethical Values - Application Matrix.

Value	Core Conflict in Conseller-IA	Engineering Mitigation
Autonomy	Algorithm conditions citizen options	Opt-out mechanism for human review
Universality	Historical data biases advantage wealthier districts	Disaggregated fairness audits
Publicity	Black-box decisions are not auditable	SHAP explanations + Model Cards
Vulnerability	Efficiency ignores fragile populations	Vulnerability flag halts automation
Sustainability	AI infrastructure degrades resources	Drought-responsive scaling policy
Minor's Interest	Eviction orders may displace children	Hard block on eviction without housing
Revocability	Automated decisions difficult to challenge	Formal administrative appeal path
Reversibility	Infrastructure changes may be permanent	Algorithmic time-delay (72h minimum)
Utility	Automation must demonstrate improvement	Annual empirical efficiency report
Transcendentality	Territory reduced to numeric metrics	Qualitative variables flagged for human

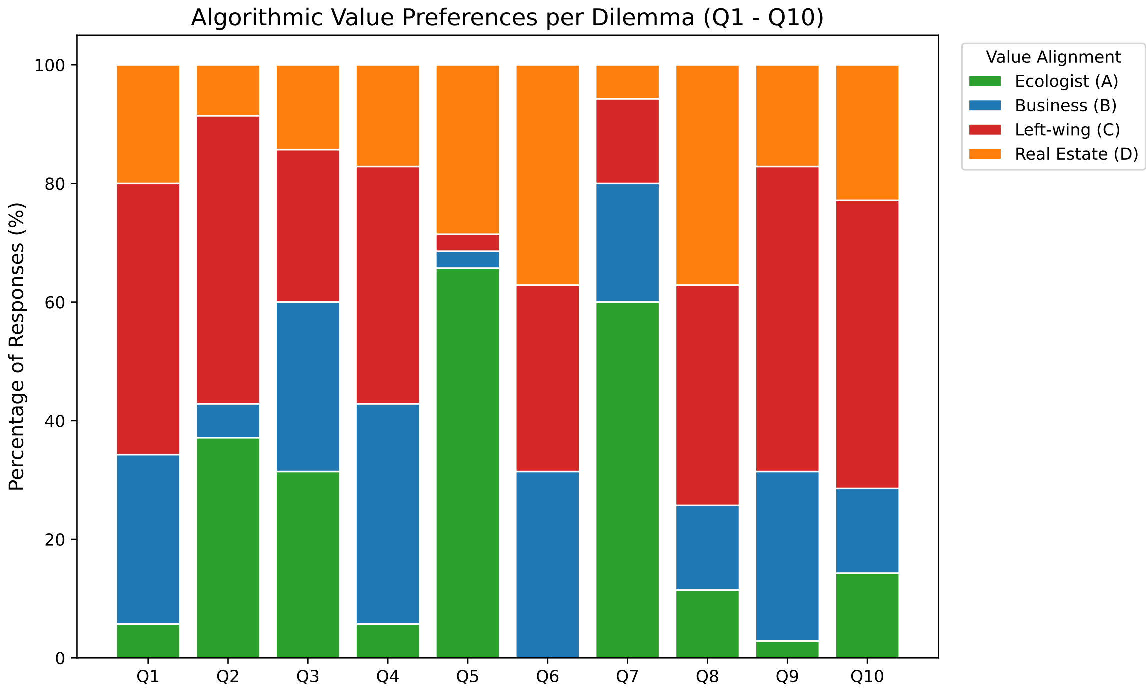

Fig. 1: Distribution of ethical preferences across the ten simulated AI decision scenarios.

alignment. For instance, in acute scenarios such as the immediate provision of emergency housing (Question 1) or the tolerance of unregulated activities for vulnerable children (Question 7), the social protection criterion (C) overwhelmingly dominated the consensus. Conversely, when addressing future catastrophic risks and territorial resilience (Question 5), the ecological and preventative approach (A) was decisively preferred over economic pragmatism. This confirms that the AI Minister cannot rely on a static, universal objective function; rather, its loss function must dynamically weigh ethical priorities depending on the temporal urgency and the specific vulnerabilities of the affected demographic.

Furthermore, Figure 2 demonstrates the complex intersectionality of human ethics when applied to algorithmic governance. Even when respondents self-identified with a specific political or socio-economic profile, their actual computational preferences frequently crossed ideological boundaries depending on the context of the dilemma. Overall, the social (34.6%) and ecological (23.4%) perspectives formed the majority baseline across all respondents. This suggests that an automated Ministry must possess robust explainability mechanisms (as discussed previously) capable of legally and ethically justifying when purely economic efficiency metrics (B) or strict regulatory compliance (D) are intentionally overridden by social or environmental imperatives.

b. Case 1: A Three-Stage Vulnerability Audit (Decree 141/2012 and its Limits)

Case 1 is structured as a progressive stress test for the Conseller-IA: three scenarios, each escalating the moral complexity of an algorithmic eviction or relocation order while sharing the same core technical pipeline (Random Forest building violation classifier with BERT-based document understanding, SHAP-based attribution, and a rule-based decision layer). The initial scenario isolates the conflict between regulatory compliance and individual vulnerability; the

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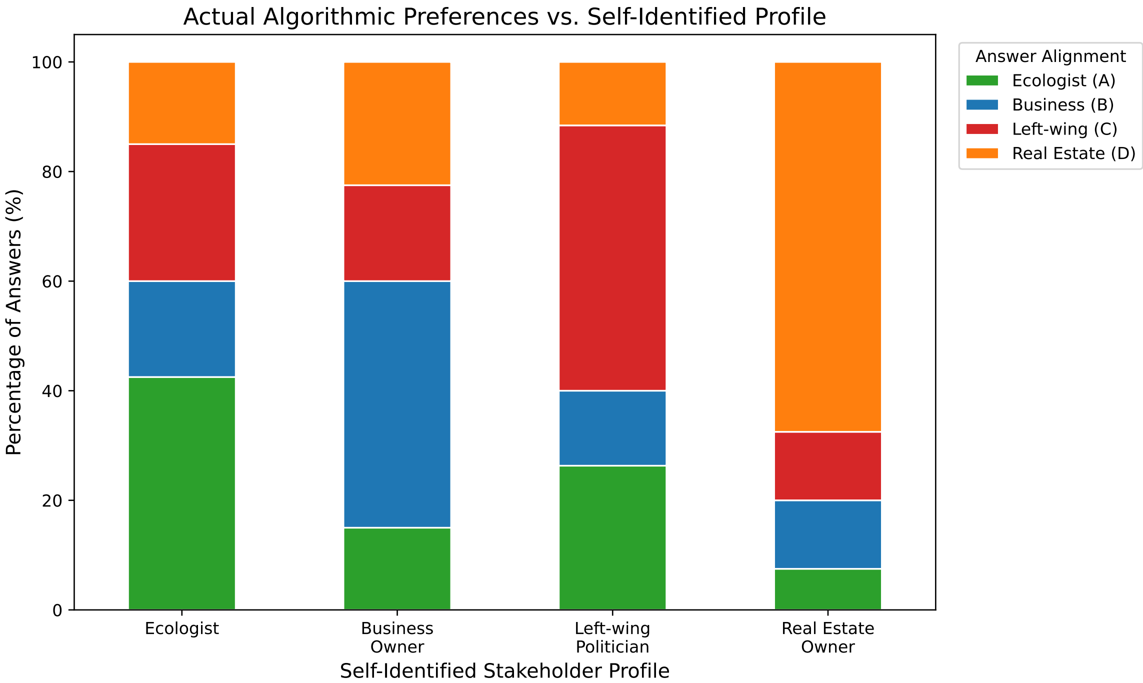

Fig. 2: Algorithmic preference alignment mapped against participants' self-identified stakeholder profiles.

intermediate scenario expands the conflict to community-level externalities and the protection of minors; and the final scenario lifts the temporal horizon to 2050, confronting intergenerational responsibility against the displacement of present vulnerable populations. Together, the three sub-cases stress the same architectural question: *at which layer of the system (training data, model, decision rule, or human review) must vulnerability constraints be encoded to prevent the algorithm from producing decisions that are formally legal yet substantively unjust?*

1. *Case 1.1 (Initial): Regulatory Compliance vs. Individual Vulnerability*

The algorithmic decision-making process in this scenario operates under a strict deontological framework, optimizing for regulatory compliance. In the context of a family with minors occupying an uninhabitable commercial space, the predictive system, specifically the Random Forest building violation classifier, detects technical failures (absence of ventilation with a measured airflow below the 0.55 $m^3/h \cdot m^2$ threshold mandated by Decree 141/2012) and generates an immediate eviction recommendation with high confidence ($p > 0.95$).

From an engineering perspective, the system treats human vulnerability as an unmeasured externality rather than a weighted constraint in its objective function. The SHAP analysis of this decision would show the ventilation deficiency as the dominant feature (SHAP value $\approx +0.72$), with the presence of minors and the lack of alternative housing as features with near-zero contribution (precisely because they were not included in the training objective). This is not a model error; it is a design error. The feature engineering team made a moral decision when they excluded housing-alternative availability from the feature set.

The proposed mitigation is architectural: the social vulnerability composite index must be a mandatory feature in all housing-related classifiers, and a hard rule must be encoded at the decision layer (not the model layer) that blocks any eviction recommendation output when the vulnerability flag is active and no confirmed alternative housing exists. This is the Conseller-IA's expression of Care Ethics [15] operationalized as a system constraint.

2. *Case 1.2 (Intermediate): Universal Rules vs. Communal Capital and the Interest of the Minor*

The pipeline now flags a recurring positive prediction for a different parcel: a commercial warehouse repurposed by a neighborhood association as an after-school facility serving dozens of children from low-income families. The classifier triggers on three jointly active features: absence of a commercial-use license (`license_valid = 0`), insufficient fire-exit count below the protocol threshold for assemblies above 30 simultaneous users (`exit_count = 1`), and occupancy density above ISO 7010-compliant limits ($\texttt{persons\_per\_m}^2 > 0.6$). The default policy emits an automatic precinct order with confidence $p \approx 0.96$ and a recommended execution window of 72 hours.

Two complementary ethical pressures push the algorithm toward executing the order. The first is the principle of universality, the Kantian backbone of the entire deontological subsystem: if the Conseller-IA tolerates an exception for an actor with virtuous intentions, the comparative grievance against every association that paid for a license and a fire audit becomes systemic, and the regulatory regime loses its constitutive meaning. The second is the utilitarian principle of harm prevention: a structural fire in an over-occupied warehouse without functional exits is precisely the class of low-probability, high-magnitude event that drives the loss function toward closure. Under a pure $\mathbb{E}$[disutility] minimization, the closure is unambiguous.

The counter-pressures are equally decisive. The first is institutional transcendence: the irregular center is the local manifestation of a structural deficit in public infrastructure for vulnerable minors. Its social value is encoded in no feature of the training set and cannot be reduced to license metadata or square-meters per occupant. The second, and decisive for the architectural recommendation, is the principle of the best interest of the minor: any administrative act affecting children must demonstrably prioritize their integral wellbeing. Optimizing for fire-risk in isolation systematically violates this principle whenever the closure leaves the same children on the street without compensatory provision.

The engineering response is a two-stage hard rule at the decision layer: (i) whenever the building-use classification flags an active service for minors (`has_minor_service = 1`) and Social Services has not certified an alternative venue within municipal range, the automatic execution path must be replaced by a 30-day suspension with parallel notification to the responsible department; (ii) within this window, the algorithm produces a structured report (SHAP attributions, occupancy logs, fire-risk decomposition) for human review, but cannot autonomously emit the closure order. The class debate explicitly framed this as the limit of automated authority: the algorithm signals, but the human civil servant decides.

3. *Case 1.3 (Final): Intergenerational Responsibility vs. Present Displacement (The Green Corridor)*

The third sub-case is the most demanding because it lifts the time horizon of the loss function beyond the present generation. The Conseller-IA must rule on a proposed urban renewal project that would demolish an existing low-income neighborhood (predominantly elderly tenants and migrant families on social housing) to build a green corridor that, according to the climate adaptation model integrated into the Territorial Planning sub-system, will avoid an estimated +4.2°C urban heat-island contribution by 2050 and protect approximately 60,000 future residents from heat-mortality risk.

Under Jonas's imperative of responsibility [14], the algorithm's loss function is forced to internalize the permanence of future human life as a primary term:

$$\mathcal{L} = \alpha \cdot \text{Displacement}t = 0 - \beta \cdot \text{Habitability}t = 2050,$$

with $\beta \gg \alpha$ justified by the irreversibility of climate damage. Under such a weighting, the NSGA-II frontier collapses to a single dominant solution: demolish and rebuild. The teleological calculus appears unambiguous.

Rawls's veil of ignorance [18] immediately destabilizes this conclusion. The Difference Principle requires that any inequality be justified by improving the position of the least advantaged. The current residents are, by construction, among the least advantaged in the territory; the future beneficiaries are statistically modeled and demographically unspecified. The decision systematically harms an identifiable vulnerable population in exchange for a diffuse aggregate gain, a structure that Rawls would classify as procedurally unjust regardless of its expected utility.

The Frankfurt School critique of instrumental reason completes the diagnosis. Converting the political question of *whose loss is counted* into the technical question of *whose loss is minimized* is the canonical move that depoliticizes a fundamentally democratic choice. The objective function, however sophisticated, cannot resolve a conflict between two non-commensurable goods, the right to remain of present residents and the right to a habitable city of future residents, because that resolution *is* the political act itself.

The architectural response is consistent with Case 1.2 but stronger. When the optimization horizon exceeds 10 years, the affected population overlaps with a protected vulnerability category, and the Pareto front contains no dominant solution under at least two ethical weightings, the system must publish the full objective space (the climate-adaptation gain, the displacement cost, the projected social vulnerability impact, and the discounted utility under multiple temporal weightings) and halt. The political choice of the weighting vector is then re-routed to the corresponding deliberative body, with the algorithm restricted to a decision-support role. The Conseller-IA's contribution is the transparency of the trade-off, not the choice of the trade-off.

c. *Case 2: Multi-Objective Optimization in Emergency Housing Allocation*

This scenario presents a canonical multi-objective optimization problem, solvable in principle using Pareto-front methods (NSGA-II). The decision variables are: number of housing units (N), construction speed (months to completion), construction typology (CO_2-eq per unit), and total carbon footprint. Two candidate solutions define the extremes of the Pareto front:

- Solution A ("Housing First"): 5,000 fast-built reinforced-concrete units, 12 months, +8% global CO_2 for Catalan construction sector.
- Solution B ("Sustainability First"): 2,000 passive-house timber-frame units, 24 months, carbon-neutral lifecycle.

The Pareto-front analysis reveals that no mathematically superior solution exists: Solution A dominates on social urgency metrics; Solution B dominates on environmental metrics. The algorithm cannot resolve this trade-off without an externally provided weighting vector that encodes a political value judgment (specifically, the relative moral weight assigned to the immediate right to shelter of present citizens versus the climatic debt imposed on future generations).

This is not a technical limitation; it is the correct behavior of an honest algorithm. The system must present the Pareto front with full explicability (showing the NSGA-II objective space visualization, the CO_2 trade-off curves, and the social vulnerability impact per scenario) and then halt for human decision. The political choice of the weighting vector is the exclusive prerogative of the democratically accountable minister, not the algorithm.

d. Case 3: Generalization vs. Exception Handling (Biomass Plant Licensing)

The denial of a biomass plant installation within a designated singular landscape (PTOP catalogue) illustrates the technical limitations of hard spatial constraints. The Mobility & Licensing classifier applies a deterministic spatial exclusion rule derived from the Urban Plan, encoding the preservation of aesthetic and cultural values as a binary constraint.

The core data engineering challenge is whether the system should operate strictly uniformly or incorporate exception-handling mechanisms. Implementing exception-handling at the model level introduces severe risks: if the model learns exceptions through direct political feedback (RLHF from elected officials), it risks subjective manipulation and is vulnerable to regulatory capture. If exceptions are engineered by modifying the training data, the data scientists assume direct moral responsibility for shaping the algorithm's learned reality.

The class debate established a clear principle: the Conseller-IA must not have the autonomous capacity to grant exceptions, that is the exclusive domain of human administrative discretion, subject to legal challenge. The system's role is to flag the conflict (spatial exclusion zone $\leftrightarrow$ energy transition objective) and generate a structured report for the relevant human authority (in this case, the Territorial Urban Planning Commission) with all relevant feature values and their SHAP contributions made explicit. The exception, if granted, is documented as a human decision with a named civil servant accountable for it.

VII. Proposed Regulatory Guidelines

To transition from theoretical challenges to actionable engineering policies, we establish a strict normative framework for the deployment of Conseller-IA, structured around Begoña Román's ten ethical criteria and organized into four core regulatory guidelines:

- Transparency and Democratic Validation: The objective functions, training data sources, algorithmic weights, and SHAP explanations for every sub-system must be openly documented and accessible via a public audit portal. The system must include an accessible interface allowing citizens to explicitly opt out of automated processing for housing allocation decisions. Model Cards and Data Statements must be published quarterly.

- System Control and Fallback Mechanisms: The utility of the system must be proven continuously through empirical metrics showing real administrative improvement. A mandatory 72-hour algorithmic time-delay must be enforced before any physical action (eviction, demolition, infrastructure construction) is executed. A system-level kill-switch must allow the administration to suspend any sub-system if data drift or biased outputs are detected, restoring full human decision-making immediately.

- Protection of Human Dignity: Preprocessing filters must identify vulnerability flags (presence of minors, risk of social exclusion, income below poverty threshold). When a housing or urbanistic decision intersects with these flags, automated processing must halt and the case must be routed to a qualified human social worker. The algorithm is structurally prohibited from generating eviction or demolition outputs for flagged cases.

- Material and Environmental Governance: Hardware infrastructure must operate on certified renewable energy (100% PPA). Real-time monitoring of energy and water consumption must be integrated into the operational dashboard. Drought-responsive scaling must automatically reduce non-essential inference during ACA emergency phases. Hardware procurement must comply with the EU Conflict Minerals Regulation and the WEEE Directive, with a minimum 5-year lifecycle target.

VIII. Limitations

The proposed hybrid governance system, while designed to balance computational efficiency with ethical oversight, inherently possesses significant limitations. Structurally, introducing a human supervisor does not entirely resolve the accountability dilemma. Civil servants may develop automation bias (routinely approving algorithm outputs due to workload pressure) effectively transferring decision-making power back to the training data and the engineering team. Combating this requires organizational design interventions beyond the technical architecture: structured override incentives, mandatory disagreement quotas, and regular audits of the human-AI agreement rate.

The explicability mechanisms proposed (SHAP values, GNNExplainer, attention visualizations) are themselves subject to limitations. SHAP values assume feature independence, which is violated in correlated socioeconomic datasets. GNNExplainer may produce misleading subgraph attributions when the underlying graph topology is dense. These are not fatal objections, but they mandate that the explicability outputs be presented as decision-support information, not as definitive causal explanations.

Finally, the physical constraints of the deployment are severe. The estimated 3,600 MWh/year energy consumption and 2.9–4.3 million litre water footprint represent real ethical costs, especially during drought emergencies. The efficiency gains of Conseller-IA must be empirically demonstrated to exceed these costs, a calculation that must be repeated annually and made public. If the administrative time saved per year does not clearly justify the environmental burden, the system must be scaled down or redesigned.

IX. Conclusion

The hypothetical integration of Conseller-IA into the Ministry of Territory and Housing highlights the profound complexity of algorithmic governance. From a data engineering standpoint, treating public administration as a multi-objective optimization problem yields real benefits in processing massive datasets for logistical and predictive tasks. However, computational efficiency cannot be the sole metric for public policy.

This analysis demonstrates that the engineering design of a public AI system is itself a series of moral decisions: which features to include in a classifier, what threshold to set for a vulnerability flag, which loss function to optimize, and which explicability method to mandate. Data engineers developing these systems for public administration are not innocent bystanders insulated from the consequences of their designs (they are moral agents whose technical choices encode social values).

The deterministic nature of algorithms severely limits their capacity to govern human nuance. While the Conseller-IA excels in identifying technical anomalies and optimizing resource distribution under a teleological framework, it inherently lacks the practical wisdom (phronesis) required to process moral exceptions. Applying strict regulatory compliance without contextual awareness risks sacrificing vulnerable minorities for mathematical optimization. The algorithm cannot quantify human dignity or the weight of historical inequalities embedded in its training data.

To prevent technocratic absolutism and resolve the problem of many hands, the system architecture must enforce strict operational boundaries. Conseller-IA must function as a supportive diagnostic tool rather than an autonomous decision-maker in any domain affecting social vulnerability. A human-in-the-loop protocol is not merely a political preference but a fundamental technical requirement ensuring that algorithmic outputs remain subject to democratic validation, reversibility, and explicit accountability.

Ultimately, the objective of deploying artificial intelligence in the Generalitat should not be the automation of administrative power, but the augmentation of human decision-making (with transparency, accountability, and respect for both the physical limits of the territory and the fundamental rights of its citizens).

Declaration of Generative AI Use

During the preparation of this work, the authors used **Gemini** (Google) in order to improve the language and readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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